

Programming Assignment 2

Objective:

In this assignment, you are provided with four images (`image1.jpeg`, `image2.jpeg`, `image3.jpeg`, and `image4.jpeg`). You are required to write a Python script using the OpenCV library to practice several image processing functions covered in `practice_04_image_processing_with_OpenCV.pdf`.

Assignment Description:

- Resize all images to 200×200 pixels.
- After resizing `image1.jpeg`, create a figure pane consisted of two rows and two columns and display the image in four formats:
 - Thresholded using `cv2.THRESHOLD_BINARY` and display it in the first row, first column.
 - Thresholded using `cv2.THRESHOLD_BINARY_INV` and display it in the first row, second column.
 - Thresholded using `cv2.THRESHOLD_TRUNC` and display it in the second row, first column.
 - Thresholded using `cv2.THRESHOLD_TOZERO` and display it in the second row, first column.
- After resizing `image2.jpeg`, create a figure pane consisted of three rows and two columns and display the image in five formats:
 - Smoothed using gamma correction using gamma value = $\frac{1}{4}$ and display it in the first row, first column.
 - Smoothed using a 2D filter using a 3×3 kernel $\begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix}$ and display it in the first row, second column.
 - Smoothed using the `blur` function using `kernel_size=(3,3)` and display it in the second row, first column.
 - Smoothed using the `GaussianBlur` function using `kernel_size=(3,3)` and the standard deviation in the X direction = 10 and display it in the second row, second column.
 - Smoothed using the `medianBlur` function using `kernel_size=3` and display it in the third row, first column.
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- After resizing `image3.jpeg` and converting in grayscale, create a figure pane consisted of two rows and two columns and display the image in five formats:
 - Display in the first row and the first column after executing the Sobel function in the X direction.
 - Display in the first row and the second column after executing the Sobel function in the Y direction.
 - Display in the second row and the first column after executing the Laplacian function in the X direction.

- After resizing `image4.jpeg`, create a figure pane consisted of two rows and two columns and display the image in five formats:
 - Display in the first row and the first column after drawing a rectangle inside the image with a red boundary.
 - Display in the first row and the second column after drawing a circle inside the image with a red boundary.
 - Display in the second row and the first column after drawing a diagonal red line inside the image.

Submission Instructions:

- Please use eLearning to submit your code. Any submission using other tools (e.g., email or MS Teams) is not accepted.